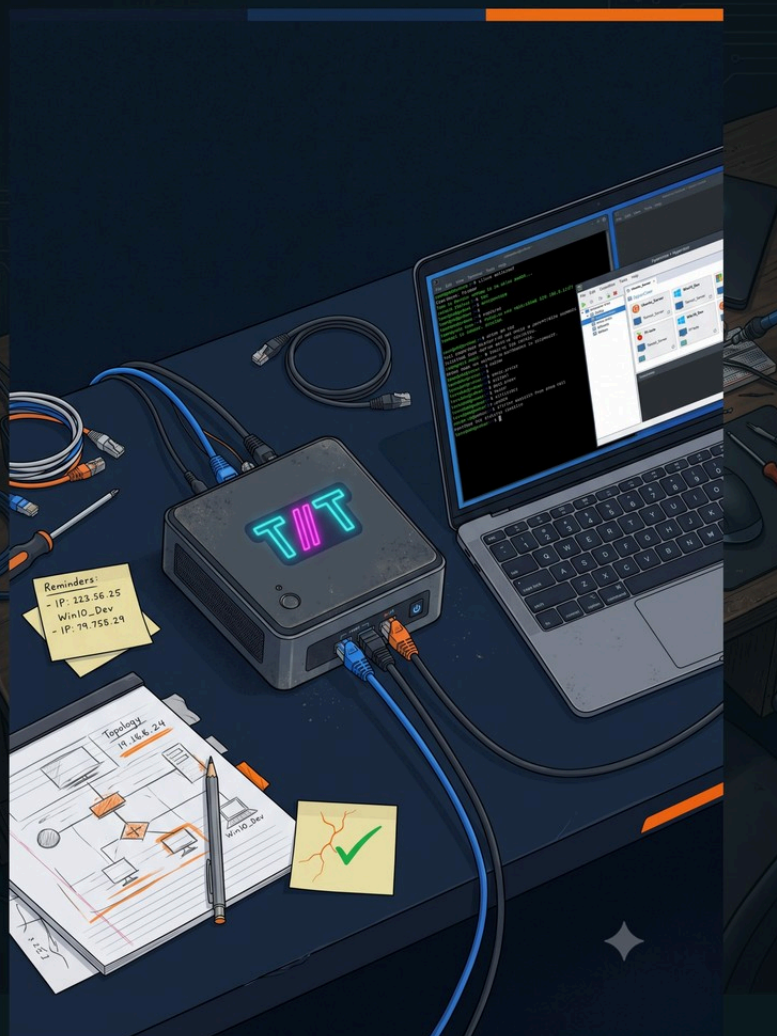


FROM CERT TO CAREER:
THE HANDS-ON IT LAB MANUAL

BUILD IT, BREAK IT

18 HANDS-ON LABS THAT TURN A CERTIFICATE
INTO COMPETENCE



BUILD IT. BREAK IT. GET HIRED.

JAMES DESTRADES JR. · TWILIGHT TECHNOLOGY

Build It, Break It – The 5-Page Roadmap

PAGE 1 – The Mindset Shift

A certificate proves you can recall. A lab proves you can do. Employers hire the second thing.

If you're holding a CompTIA A+ and a growing sense that it hasn't opened the door you were promised — this is for you. So is the person who hasn't taken a single exam yet and doesn't know where to begin. Same doorway. You just walk through it at different speeds.

Here's the uncomfortable truth the cert won't tell you: knowing what a DHCP scope *is* and having watched a client pull an address from one *you built* are two different kinds of knowledge. The first lives in short-term memory until exam day. The second lives in your hands. When a real machine fails to get an address at 2 p.m. on a Tuesday with a user over your shoulder, only one of those shows up to help you.

THE PROMISE — Work through the projects and you won't say *"I studied networking."* You'll say *"I built a domain, joined clients to it, pushed policy, then broke DNS on purpose and walked the diagnosis from the bottom up."* One of those sentences gets a second interview.

You don't need permission to start. You don't need a job to start. You need a machine you already own and the willingness to break things where breaking them is free.

PAGE 2 – Bust the Cost Myth

The barrier was never money. It was knowing where to start.

The #1 reason people never build a home lab is a picture in their head: a server rack humming in a basement, a power bill to match. That's fiction for a beginner. A home lab is the laptop you're reading this on.

Tier	What it is	Reality check
Tier 0	The laptop/desktop you already own (8 GB starts, 16 GB comfortable)	\$0. Runs 1–2 VMs. Enough to start today.
Tier 1	A used mini PC — OptiPlex, EliteDesk, ThinkCentre Tiny	\$80–150 used. The sweet spot for a dedicated lab.
Tier 2	Mini PC / workstation, 32–64 GB RAM	Only if you catch the bug. Runs a whole domain at once.

THE HONEST COMPARISON — Imagined rig: ~\$2,000. The machine that teaches you the same fundamentals: the one you already own, or a ~\$100 office PC. For a beginner the learning delta rounds to nothing. Start now.

Everything is free and legal: Windows Server 180-day eval ISOs · Windows client dev VMs · VirtualBox / Hyper-V / Proxmox · Ubuntu / Rocky / Debian · pfSense / OPNsense · a free Microsoft 365 developer tenant · osTicket · Wazuh. No piracy, ever.

PAGE 3 — Why Labs Beat Cramming

Labs teach you the layer beneath the symptom. In IT — especially at an MSP — your job is to troubleshoot other people's systems. The fast, trusted techs aren't the ones who memorized the most facts. They're the ones who understand systems bottom-up, so they know *which layer to suspect first*.

REAL TALK — the corrupted cache. A user says an app won't load. The crammer uninstalls, reinstalls, reboots, hopes — 40 minutes. The lab-trained tech knows the app is a front end on a file structure, finds the one corrupted cache file under AppData, deletes it; the app reopens and resyncs clean. Two minutes. Same ticket. In an MSP that difference is billable efficiency — and it's what gets you promoted.

A cert hands you the OSI model as a list to memorize. A lab gives you the *instinct* to know, when someone says "it's not working," whether to suspect the app, the file under it, the permission on that file, the network path, or name resolution three steps back. You can't read that instinct into your hands. You earn it by building the thing and breaking it.

PAGE 4 — The 18-Project Map

Five tracks. Each project builds on the last. Each one is a story you'll tell in an interview.

 **Windows (7)** — the backbone, empty hypervisor → working domain **1** Hyper-V foundation · **2** promote a domain controller · **3** DNS + DHCP · **4** join a client + build OUs · **5** Group Policy · **6** file

shares + permissions · 7 print server + GPO-deployed printers

☁ **Cloud & Identity (1)** — the cloud half of the modern job, no hardware 8 build & break a Microsoft 365 + Entra tenant — users, licensing, MFA + Conditional Access, the AADSTS50105 fix

⚙ **MSP Operations (2)** — run it like a managed shop 9 stand up osTicket · 10 central patching (WSUS or RMM)

🐧 **Linux (6)** — your essential second language 11 Linux lab foundation · 12 the filesystem hierarchy · 13 users/groups/permissions · 14 shell fundamentals · 15 SSH + remote admin · 16 logs + Samba (talking back to Windows)

🎯 **Capstones (2)** — break what you built 17 network troubleshooting from the bottom up · 18 attack & defend your own domain with a SIEM watching

Every project ends with a **document-it** step. That documentation is the portfolio.

PAGE 5 — Turn Labs Into a Job

Unseen skill doesn't get hired. Here's how the labs become an offer letter:

1. **Document everything as runbooks.** Each project, written up step-by-step — the exact skill the job needs, and the raw material for your portfolio.
2. **Put it where people can see it.** A public GitHub repo titled *Home Lab Projects*, a folder per project with notes, diagrams, screenshots. More convincing than any résumé line.
3. **Talk about it in interviews.** *"How would you troubleshoot a user who can't get on the network?"* becomes *"In my lab I broke DHCP and DNS separately so I could tell their symptoms apart — here's how I'd work it bottom-up."* That sounds like experience because it **is** experience.

THE WHOLE POINT — Nobody's permission is required to start building competence today. You don't need a job to gain experience — the lab *is* the experience, and the documentation is the proof. The barrier was never money. It was knowing where to start. Now you know.

→ GET THE FULL MANUAL

This roadmap is the map. **Build It, Break It (full edition)** is the territory — all 18 projects with every build step, every break-it-then-fix-it drill, and the portability notes that make the skills transfer anywhere.

Build it. Break it. Get hired. → *[full edition link]* · Want a personal game plan to go with it? *[/strategy — \$99 Cert Strategy Session]*

Part of From Cert to Career — Pass the exam · Work the job · Prove you can do it.